UBICOMP ISWC2025

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Reality Modeling Language (RML)

Mikel Salazar, Ph.D.
VR Department. IFE
Halden, Norway

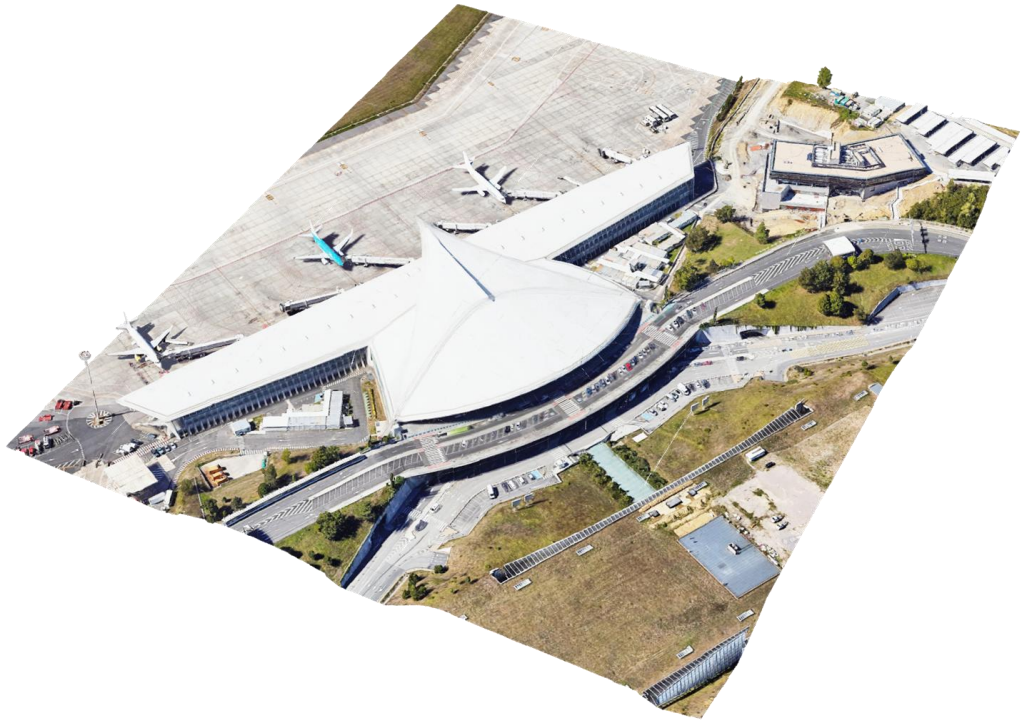
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The Many Different Ways to Understand Reality

Due to the extreme complexity of the world we live in, we all operate with simplified mental models of it. And we subconsciously ignore the parts that are not meaningful to us.

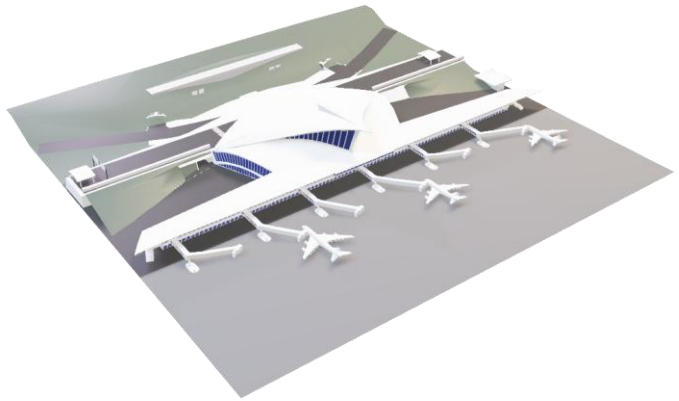
This often results in surprisingly different understandings the real world. And even when we use the same terms, the meaning behind them often have slight differences.

This is a serious problem when creating digital versions of those mental models, because those seemingly **small dissimilarities result in incompatible file formats, loss of critical information during conversions, etc.**



The Limitations of Current Approaches

Rendering (SceneGraph)

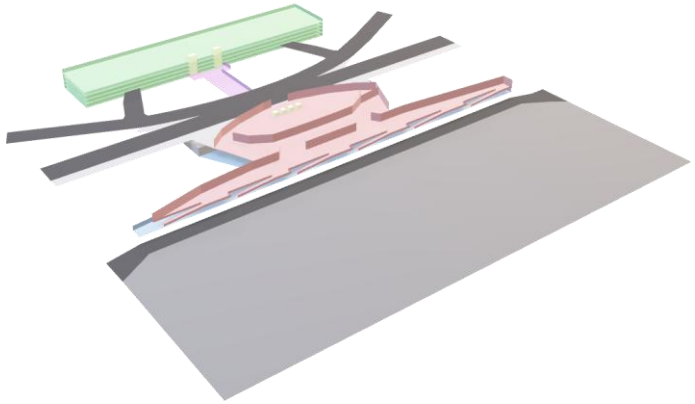


Good 3D Visual Data

Basic Semantic Data

No Spatial Data

Architecture (BIM-IFC)

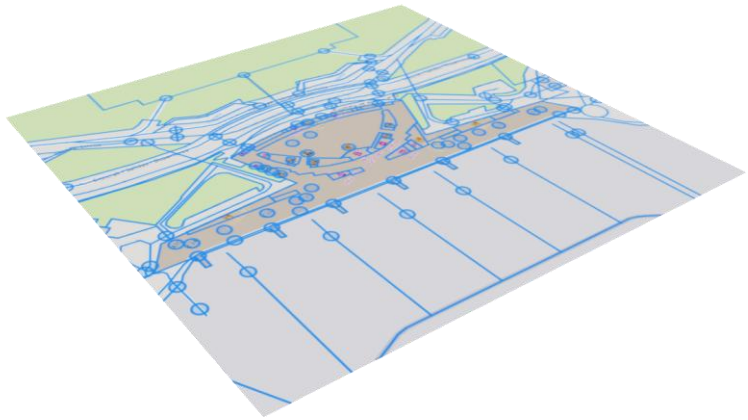


Simple 3D Visual Data

Good Semantic Data

Simple Spatial Data

GeoSpatial (OSM)



No 3D Visual Data

Basic Semantic Data

Good Spatial Data

Modeling Reality With a Description Language

The Reality Modeling Language (RML) is a formal description language that provides a simple but extensible and interoperable way to specify reality layers (i.e., “verses”) and their entities, whether they are real/physical or virtual/synthetic in nature.

In practice, **it provides a way to easily describe physical environments**, so that developers can build interaction spaces where users (humans and robotic systems) can collaborate to achieve complex tasks.

Although it can be considered to something similar to a ”Digital Twins Definition Language”, RML is a more encompassing solution. One that **aims to generate an interoperability layer upon which it will be possible to build a continuous, federated and XR-based interaction paradigm.**

RML **is platform independent and format agnostic description language**, so it can be written in JSON, XML, YAML or any other human-readable format.

More than (Yet Another) 3D Model Format

Just like how HTML doesn't use in a webpage (just describes where they go), **RML doesn't try to redefine how the 3D models are rendered,** but how they are placed in the real world and the spaces they generate.

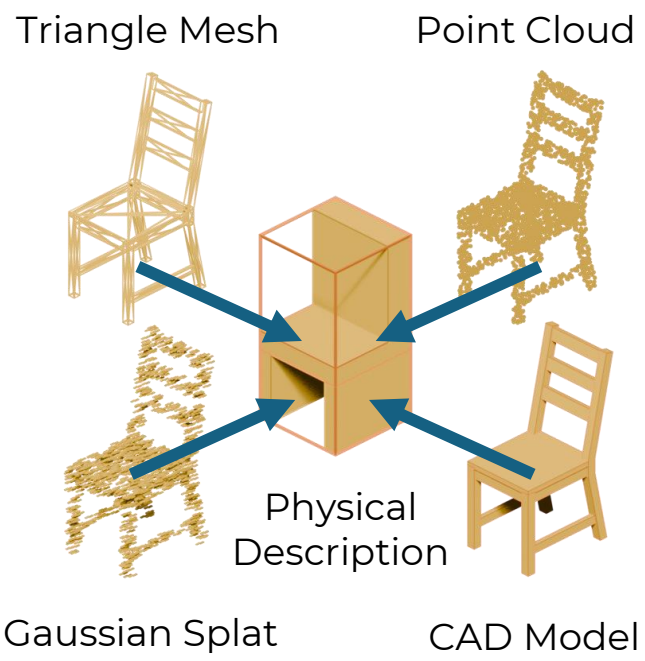
While RML doesn't include any logic systems to avoid overcomplicating the syntax (and any implication), **it is a great foundation over which to build all kinds of Spatial Computing experiences.**

The first version of the Language was created within the OSCAR4US EU Project, but as part of OpenARCloud and the MSF, **we are already talking about how to standardize RML and integrate it within other standards.**



The Main Design Principles of RML

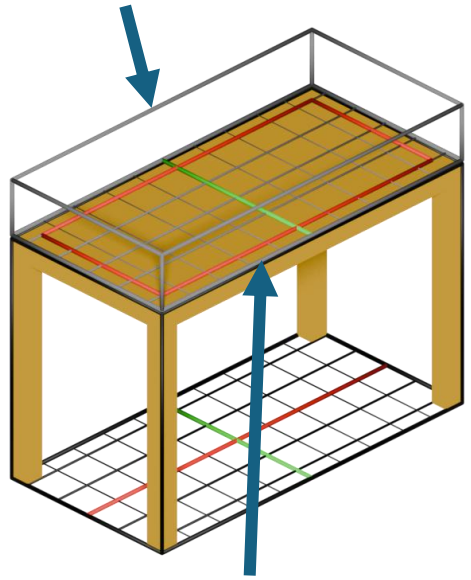
Multiple Representations, One Single Description



Also, the spatial data can be reused to generate “blocky” representation.

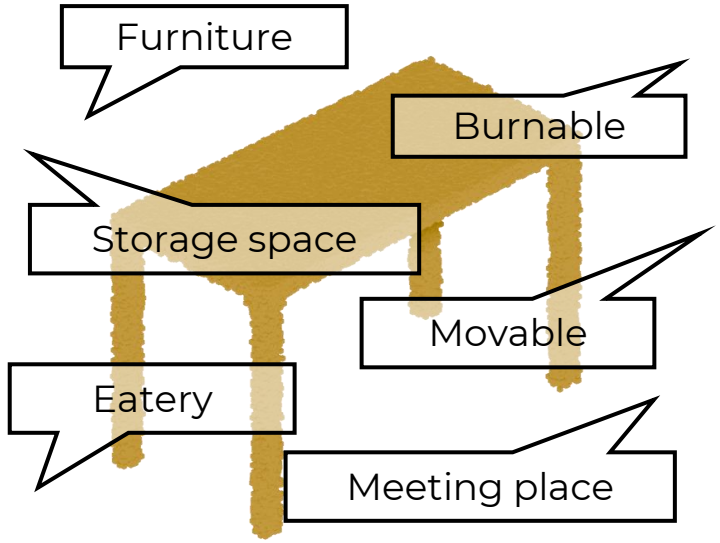
Spatial Information (and Location/Distribution data)

The “top the table” space can be auto generated



Special coordinate systems can be defined to distribute linked entities

Extensible Semantic System (from different ontologies)



Complementary understandings of the same entity can coexist.

Advantages of This Approach

Interoperability: Being able to work directly (i.e. without conversion) with the same 3D data used by architects, CAD modelers, roboticists, geospatial engineers… is important. Not only for XR applications, but for anything that has a presence in the real world.

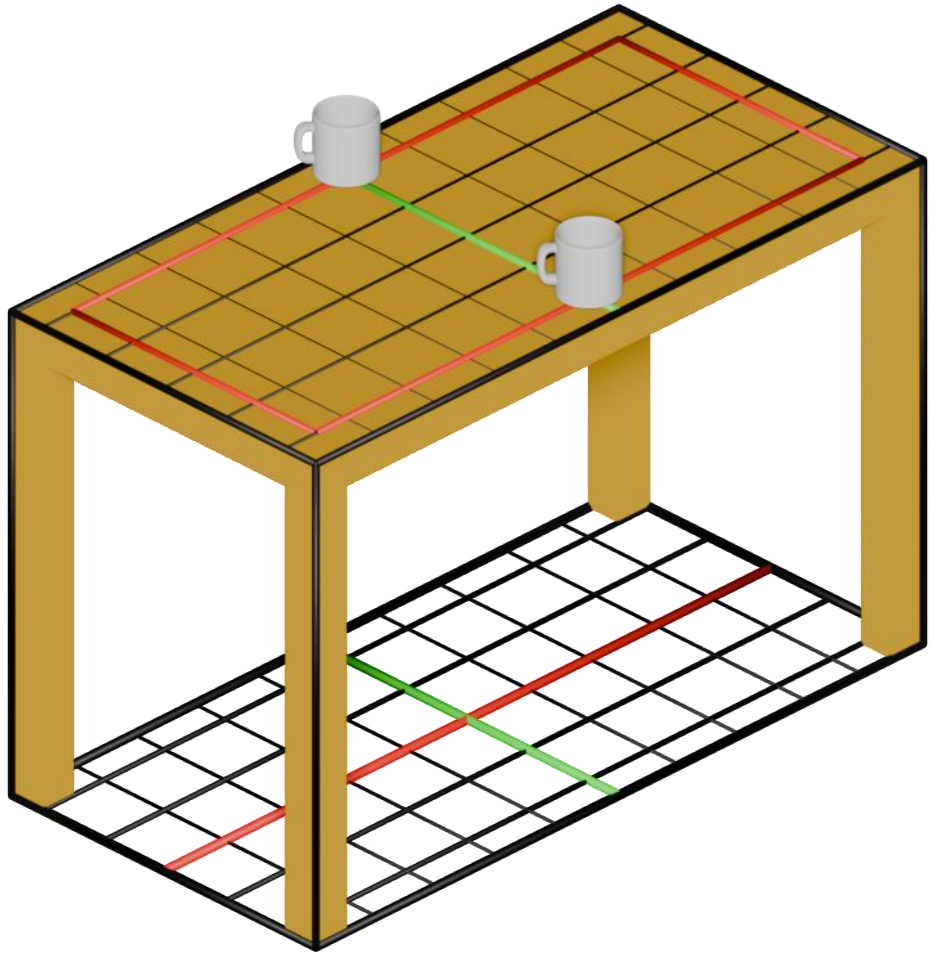
Simplicity: The relatively simple syntax of the facilitates the adoption of the language by third parties and the creation of third-party importers/exporters.

“Cooperative”: By keeping the language focused on just describing the world (without adding business logic or user interaction elements) RML is not only extensible, but also easily ***integrable into other formats and standards.***

Solves common issues: No longer will a 3D designer link a cup to a table and have to bring it up from the origin of coordinates of the latter. Physics simulation-based problems are minimized because of the spatial data. And most important… we can now actually define spaces (for the camera frustrum, collisions, navigation, etc.)

A Simple Example: Cups on a Table

```
<rml name= “SimpleExample”>
        <verse id=“example”>
                <entity id=“table”>
                        <representation
path=“table.glb”/>              <space type=“box”
                                        width=“0.5”
height=“0.65” depth=“1”/>       <space id=“top” type=“polar”
                                        width=“0.5”
depth=“1” margin=“0.05”/>       </entity>
                <entity id=“cup” location=“table.top”>
                        <representation path=“cup.glb”/>
                </entity>
                <entity id=“cup2” location=“table.top”>
                        <representation path=“cup.glb”/>
                </entity>
        </verse>
</rml>
```

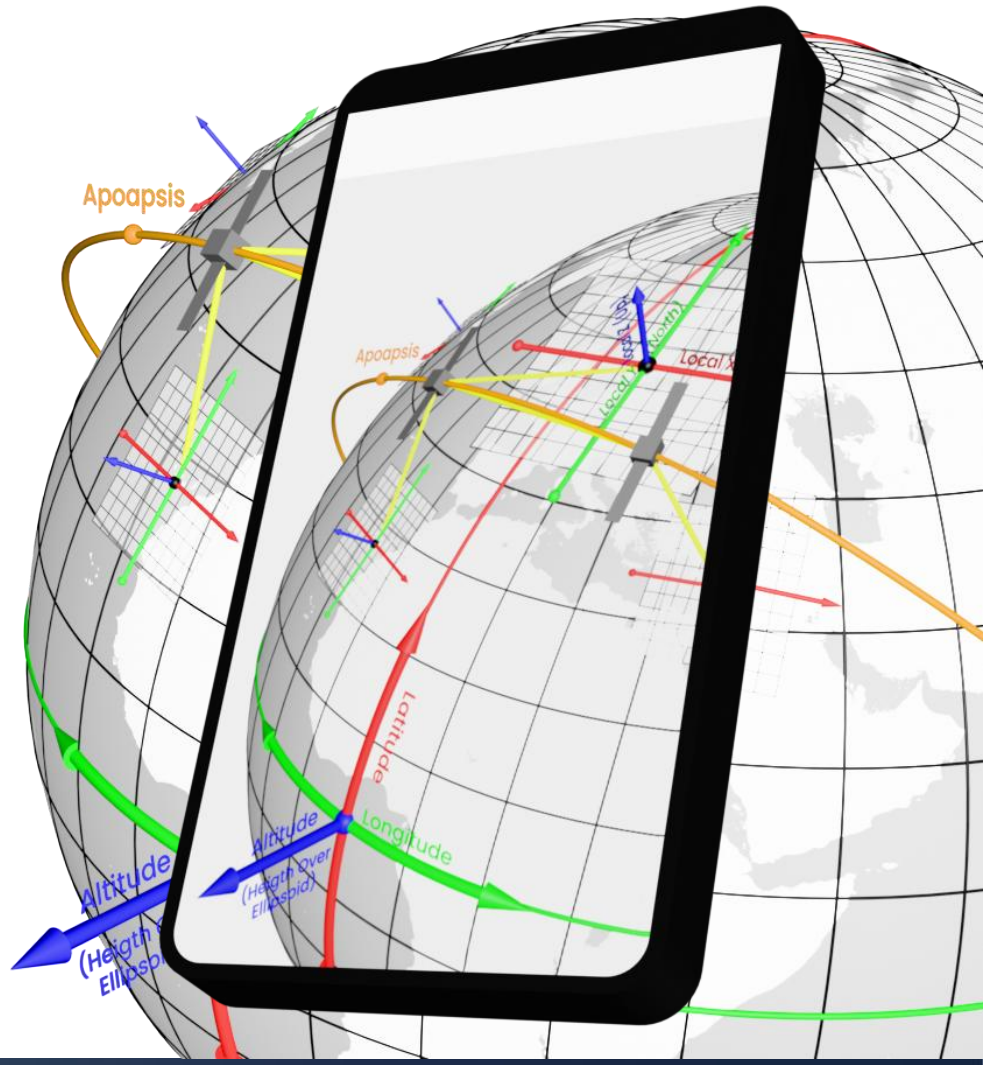


Taking over the World (Geospatially speaking)

Yet, this focus on spatial understanding is not limited to Euclidean topologies. **In RML, spaces can be geographical or orbital in nature** (and entities can be positioned within the associated coordinate reference systems).

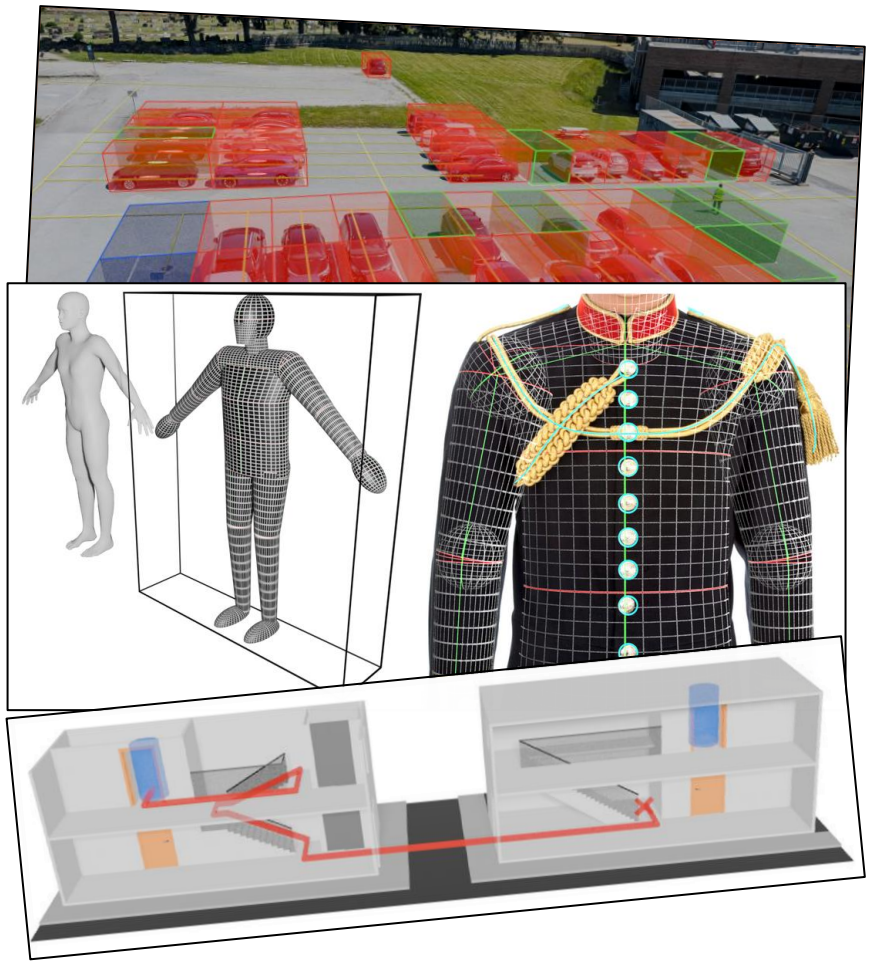
This not **only allows the creation of digital twins at an (inter)planetary scale** but also enables the modelling of satellite communications systems and celestial bodies (including the light coming from the Sun and reflected on the Moon).

Furthermore, this opens the door to the creation of **a planet-wide digital ecosystem where agents can exchange geospatial information and, even, geolocated user experiences.**



Applications of This Transversal Technology

- **Advanced Digital Twins:** The most direct application of this language is the definition of Digital Twins that take advantage of the (geo)spatial and semantic information to gain a better understanding of the environment.
- **Human Factors:** The spatial information can also describe the users themselves, enabling better human factor studies and the creation of advanced avatars (with complex clothing).
- **Spatial Navigation:** Since we have the definition of the spaces and their connections (portals), it is possible to navigate between them more efficiently (avoiding collisions with other users and hazards in the environment).



What Do You Want to Use RML for?

This technology can be used in many use cases across different industries, but we are looking for projects to turn the current prototype (TRL6) into reality (TRL9).



Relevant Links

Reality Modeling Language Sandbox:

https://github.com/MikelSalazar/RealityModelingLanguage

For direct questions: (Mikel Salazar, Ph.D.)
contact@mikelsalazar.com

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